

EXTENSION 1

Stage 6

Calculus (Ext1), C3 Applications of Calculus (Y12), Differential Equations (Ext1)

Applications of Differential Equations

Teacher: Nardene Dodd

Exam Equivalent Time: 84 minutes (based on allocation of 1.5 minutes per mark)

Questions

1. Calculus, EXT1 C3 2020 HSC 12e

Find the curve which satisfies the differential equation $\frac{dy}{dx} = -\frac{x}{y}$ and passes through the point **(1, 0)**.
(3 marks)

2. Calculus, EXT1 C3 EQ-Bank 6

The population of Myna birds in a national park is decreasing at a rate proportional to the population at that time.

- Write a differential equation that describes the situation. (1 mark)
- If the population was originally 1300 and decreased to 1040 after 5 years, find the expected population after 10 years. (3 marks)

3. Calculus, EXT1 C3 SM-Bank 5

Bacteria are spreading over a Petri dish at a rate modelled by the differential equation

$$\frac{dP}{dt} = \frac{P}{2}(1 - P), \quad 0 < P < 1$$

where P is the proportion of the dish covered after t hours.

Given $\frac{2}{P(1-P)} = \frac{2}{P} + \frac{2}{1-P}$,

- Show by integration that $\frac{t-c}{2} = \log_e\left(\frac{P}{1-P}\right)$, where c is a constant of integration. (2 marks)
- If half of the Petri dish is covered by the bacteria at $t = 0$, express P in terms of t . (2 marks)

4. Calculus, EXT1 C3 EQ-Bank 4

A varroa virus is infecting commercial beehives in a regional NSW town.

All infected hives detected so far lie within a circular region with radius 16 km and researchers believe that the increase of the radius r km can be modelled by a differential equation, where $\frac{dr}{dt} = \frac{2}{5}\sqrt{r}$ where t denotes the time in months.

What does this model predict for the radius of the region affected by the pest after t months? (3 marks)

5. Calculus, EXT1 C3 EQ-Bank 7

The rate of weekly sales of a product that has just been released can be modelled by the following differential equation

$$S'(t) = \frac{500}{(t+1)^3} - \frac{250}{(t+1)^2}$$

where S is the number of sales in thousands and t is the number of weeks since launch.

Good quality products will result in increasing sales.

- Find the function that describes the weekly sales. (2 marks)
- By calculating the number of sales for the first and fifth week, comment on the results of the launch with respect to the given information. (2 marks)

6. Calculus, EXT1 C3 EQ-Bank 8

A researcher estimates the number of brumbies in a National Park after t years can be modelled by the equation

$$B(t) = \frac{18\,000}{1 + 4e^{-t}}$$

- Sketch the function $B(t)$ over the first four years of the research. (2 marks)
- Calculate when the brumby population should reach 13 000, giving your answer to 2 decimal places. (1 mark)
- Show that $B'(t) = \frac{72\,000e^t}{(e^t + 4)^2}$ (2 marks)
- What is the maximum growth rate of the brumby population? (3 marks)

7. Calculus, EXT1 C3 2021 SPEC1 7

The velocity of a particle satisfies the differential equation $\frac{dx}{dt} = x \sin(t)$, where x centimetres is its displacement relative to a fixed point O at time t seconds.

Initially, the displacement of the particle is 1 cm.

- Find an expression for x in terms of t . (3 marks)
- Find the maximum displacement of the particle and the times at which this occurs. (2 marks)

8. Calculus, EXT1 C3 EQ-Bank 10

A tank initially contains 300 grams of salt that is dissolved in 50 L of water. A solution containing 15 grams of salt per litre of water is poured into the tank at a rate of 2 litres per minute and the mixture in the tank is kept well stirred.

At the same time, 2 litres of the mixture flows out of the tank per minute.

- i. Find the differential equation, $\frac{dm}{dt}$, where m represents the mass, in grams, of salt in the tank at time t minutes. (2 marks)
- ii. Express m in terms of t . (3 marks)
- iii. Find the concentration of salt in the liquid in the longer term. (1 mark)

9. Calculus, EXT1 C3 EQ-Bank 9

The rate of fuel running out of a leaking tank can be modelled by the following equation

$\frac{dh}{dt} = -Ae^{-0.2t}$ where h is the height of the fuel in the tank after t hours?

Initially, the height of the fuel in the tank is 4 metres and after 1.5 hours, it has fallen to 3 metres.

At what height of fuel in the tank will it eventually stabilise, giving your answer to the nearest centimetre? (4 marks)

10. Calculus, EXT1 C3 EQ-Bank 11

The maximum speed that a hammer can reach after falling vertically from the top of a skyscraper is $u \text{ ms}^{-1}$.

The hammer's speed, $v \text{ ms}^{-1}$, after falling x metres, is given by the differential equation

$\frac{dv}{dx} = c \frac{u^2 - v^2}{v}$ where c is a positive constant.

Find an expression of v in terms of x . (4 marks)

11. Calculus, EXT1 C3 2021 HSC 14b

In a certain country, the population of deer was estimated in 1980 to be 150 000.

The population growth is given by the logistic equation $\frac{dP}{dt} = 0.1P\left(\frac{C - P}{C}\right)$ where t is the number of years after 1980 and C is the carrying capacity.

In the year 2000, the population of deer was estimated to be 600 000.

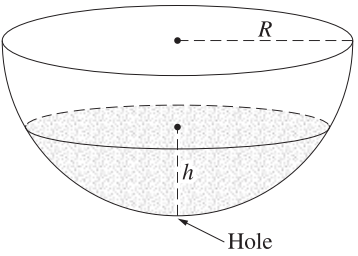
Use the fact that $\frac{C}{P(C - P)} = \frac{1}{P} + \frac{1}{C - P}$ to show that the carrying capacity is approximately 1 130 000. (4 marks)

12. Calculus, EXT1 C3 2023 HSC 13a

A hemispherical water tank has radius R cm. The tank has a hole at the bottom which allows water to drain out.

Initially the tank is empty. Water is poured into the tank at a constant rate of $2kR \text{ cm}^3 \text{ s}^{-1}$, where k is a positive constant.

After t seconds, the height of the water in the tank is h cm, as shown in the diagram, and the volume of water in the tank is $V \text{ cm}^3$.



It is known that $V = \pi\left(Rh^2 - \frac{h^3}{3}\right)$. (Do NOT prove this.)

While water flows into the tank and also drains out of the bottom, the rate of change of the volume of water in the tank is given by $\frac{dV}{dt} = k(2R - h)$.

- i. Show that $\frac{dh}{dt} = \frac{k}{\pi h}$. (2 marks)
- ii. Show that the tank is full of water after $T = \frac{\pi R^2}{2k}$ seconds. (2 marks)
- iii. The instant the tank is full, water stops flowing into the tank, but it continues to drain out of the hole at the bottom as before.
Show that the tank takes 3 times as long to empty as it did to fill. (3 marks)

Worked Solutions

1. Calculus, EXT1 C3 2020 HSC 12e

$$\frac{dy}{dx} = -\frac{x}{y}$$

$$\int y \, dy = -\int x \, dx$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + c$$

Curve passes through (1, 0):

$$0 = -\frac{1}{2} + c$$

$$c = \frac{1}{2}$$

$$\frac{y^2}{2} = -\frac{x^2}{2} + \frac{1}{2}$$

$$y^2 = -x^2 + 1$$

$$\therefore x^2 + y^2 = 1$$

COMMENT: Note the answer requires a curve equation, not a function.

Worked Solutions

2. Calculus, EXT1 C3 EQ-Bank 6

i. Let $\frac{dN}{dt}$ = rate of change of bird population at time t years.

$$\frac{dN}{dt} \propto N$$

$$\frac{dN}{dt} = -kN \quad (k > 0, \text{ decreasing population})$$

ii.
$$\frac{dN}{dt} = -kN$$

$$\frac{1}{N} \cdot \frac{dN}{dt} = -k$$

$$\int \frac{1}{N} \, dN = -\int k \, dt$$

$$\ln|N| = -kt + c$$

$$N = e^{-kt+c}$$

$$= e^{-kt} \cdot e^c$$

$$= Ae^{-kt} \quad (\text{where } A = e^c)$$

When $t = 0$, $N = 1300$:

$$1300 = Ae^0 \Rightarrow A = 1300$$

$$N = 1300e^{-kt}$$

When $t = 5$, $N = 1040$:

$$1040 = 1300e^{-5k}$$

$$e^{-5k} = \frac{1040}{1300}$$

$$-5k = \ln(0.8)$$

$$k = -\frac{\ln(0.8)}{5}$$

$$= 0.04462\dots$$

Find N when $t = 10$:

$$N = 1300e^{-0.04462 \times 10}$$

$$= 832 \text{ myna birds}$$

3. Calculus, EXT1 C3 SM-Bank 5

a. $\frac{dt}{dP} = \frac{2}{P(1-P)} = \frac{2}{P} + \frac{2}{1-P}$

$$t = \int \frac{2}{P} + \frac{2}{1-P} dP$$

$$= 2\log_e|P| - 2\log_e|1-P| + c$$

$$\frac{t-c}{2} = \log_e|P| - \log_e|1-P|$$

$$= \log_e \left| \frac{P}{1-P} \right|$$

$$= \log_e \left(\frac{P}{1-P} \right)$$

Since $0 < P < 1$: $|P| = P$ and $|1-P| = 1-P$

b. When $t = 0$, $P = 0.5$

$$\frac{-c}{2} = \log_e \left(\frac{0.5}{0.5} \right)$$

$$c = \log_e(1)$$

$$= 0$$

$$\frac{t}{2} = \ln \left(\frac{P}{1-P} \right)$$

$$e^{\frac{t}{2}} = \frac{P}{1-P}$$

$$e^{\frac{t}{2}}(1-P) = P$$

$$e^{\frac{t}{2}} - Pe^{\frac{t}{2}} = P$$

$$e^{\frac{t}{2}} = P(1 + e^{\frac{t}{2}})$$

$$\therefore P = \frac{e^{\frac{t}{2}}}{1 + e^{\frac{t}{2}}}$$

4. Calculus, EXT1 C3 EQ-Bank 4

$$\frac{dr}{dt} = \frac{2}{5}\sqrt{r}$$

$$\frac{5}{2\sqrt{r}} dr = 1 dt$$

$$\frac{5}{2} r^{-\frac{1}{2}} dr = 1 dt$$

$$\frac{5}{2} \int r^{-\frac{1}{2}} dr = \int 1 dt$$

$$5r^{\frac{1}{2}} = t + C$$

When $t = 0$, $r = 16$

$$5\sqrt{16} = C \Rightarrow C = 20$$

$$5r^{\frac{1}{2}} = t + 20$$

$$r^{\frac{1}{2}} = \frac{t+20}{5}$$

$$\therefore r = \left(\frac{t+20}{5} \right)^2$$

5. Calculus, EXT1 C3 EQ-Bank 7

$$\begin{aligned} \text{i. } S'(t) &= \frac{500}{(t+1)^3} - \frac{250}{(t+1)^2} \\ S(t) &= \int \frac{500}{(t+1)^3} - \frac{250}{(t+1)^2} dt \\ &= \int 500(t+1)^{-3} - 250(t+1)^{-2} dt \\ &= \frac{500(t+1)^{-2}}{-2} - \frac{250(t+1)^{-1}}{-1} + c \\ &= \frac{250}{t+1} - \frac{250}{(t+1)^2} + c \end{aligned}$$

When $t = 0, S(t) = 0$:

$$0 = \frac{250}{1} - \frac{250}{1^2} + c \Rightarrow c = 0$$

$$\therefore S(t) = \frac{250}{t+1} - \frac{250}{(t+1)^2}$$

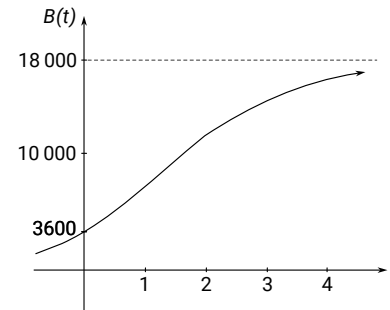
$$\begin{aligned} \text{ii. } S(1) &= \frac{250}{1+1} - \frac{250}{(1+1)^2} = 62.5 \\ \Rightarrow \text{Sales in week 1} &= 62\,500 \end{aligned}$$

$$\begin{aligned} S(5) &= \frac{250}{1+5} - \frac{250}{(1+5)^2} = 34.7222 \\ \Rightarrow \text{Sales in week 5} &= 34\,722 \end{aligned}$$

Sales drop significantly between week 1 and week 5, suggesting the product is of poor quality.

6. Calculus, EXT1 C3 EQ-Bank 8

i.



ii. Find t when $B(t) = 13\,000$

$$13\,000 = \frac{18\,000}{1 + 4e^{-t}}$$

$$13\,000(1 + 4e^{-t}) = 18\,000$$

$$1 + 4e^{-t} = \frac{18}{13}$$

$$4e^{-t} = \frac{5}{13}$$

$$e^{-t} = \frac{5}{52}$$

$$-t = \ln\left(\frac{5}{52}\right)$$

$$t = 2.34 \text{ years (2 d.p.)}$$

$$\text{iii. Show } B'(t) = \frac{72\,000e^t}{(e^t + 4)^2}$$

$$B(t) = 18\,000(1 + 4e^{-t})^{-1}$$

$$B'(t) = -1 \cdot -1 \cdot 4e^{-t} \cdot 18\,000(1 + 4e^{-t})^{-2}$$

$$= \frac{72\,000}{e^t(1 + 4e^{-t})^2}$$

$$= \frac{72\,000}{e^t\left(1 + \frac{4}{e^t}\right)^2}$$

$$= \frac{72\,000}{e^t\left(\frac{e^t+4}{e^t}\right)^2}$$

$$= \frac{72\,000}{\frac{e^t}{(e^t)^2} \cdot (e^t + 4)^2}$$

$$= \frac{72\,000e^t}{(e^t + 4)^2} \dots \text{as required}$$

iv. $B'(t) = 72\,000e^t(e^t + 4)^{-2}$

Using product rule:

$$\begin{aligned} B''(t) &= 72\,000e^t(e^t + 4)^{-2} + (-2e^t)(e^t + 4)^{-3}72\,000e^t \\ &= 72\,000e^t\left(\frac{1}{(e^t + 4)^2} - \frac{2e^t}{(e^t + 4)^3}\right) \\ &= 72\,000e^t\left(\frac{e^t + 4 - 2e^t}{(e^t + 4)^3}\right) \\ &= 72\,000e^t\left(\frac{4 - e^t}{(e^t + 4)^3}\right) \end{aligned}$$

Find t when $B''(t) = 0$:

$$\begin{aligned} 4 - e^t &= 0 \\ e^t &= 4 \\ t &= \ln 4 \\ &= 1.386\dots \text{ years} \end{aligned}$$

Checking concavity changes:

Since $e^t > 0$, $(e^t + 4)^3 > 0$ for all t :

$$\text{At } t = 1, 4 - e^1 = 1.28 > 0 \Rightarrow B''(1) > 0$$

$$\text{At } t = 2, 4 - e^2 = -3.4 < 0 \Rightarrow B''(2) < 0$$

$B'(\ln 4) =$ Max growth rate

$$\begin{aligned} B'(\ln 4) &= \frac{72\,000e^{\ln 4}}{(e^{\ln 4} + 4)^2} \\ &= \frac{72\,000 \times 4}{(4 + 4)^2} \\ &= 4500 \text{ brumbies per year} \end{aligned}$$

7. Calculus, EXT1 C3 2021 SPEC1 7

a. $\frac{dx}{dt} = x \sin(t)$

$$\begin{aligned} \int \frac{1}{x} dx &= \int \sin(t) dt \\ \log_e x &= -\cos(t) + c \end{aligned}$$

When $t = 0, x = 1$

$$\log_e 1 = -\cos 0 + c$$

$$c = 1$$

$$\log_e x = -\cos(t) + 1$$

$$\therefore x = e^{1-\cos(t)}$$

b. $x = e^{1-\cos(t)}$

$$\frac{dx}{dt} = \sin(t) \cdot e^{1-\cos(t)}$$

Find t when $\frac{dx}{dt} = 0$:

$$e^{1-\cos(t)} \neq 0$$

$$\sin(t) = 0 \text{ when } t = 0, \pi, 2\pi, \dots$$

$$x_{\max} = e^2 \text{ when } t = \pi, 3\pi, 5\pi, \dots$$

$$\text{or } t = (2k + 1)\pi \text{ for integral } k = 0, 1, 2, \dots$$

8. Calculus, EXT1 C3 EQ-Bank 10

i. $V = 50 \text{ L}$

$$\frac{dm}{dt}(\text{in}) = 15 \times 2 = 30 \text{ g/min}$$

$$\frac{dm}{dt}(\text{out}) = 2 \times \frac{m}{50} = \frac{m}{25} \text{ g/min}$$

$$\therefore \frac{dm}{dt} = 30 - \frac{m}{25}$$

ii. $\frac{dm}{dt} = \frac{750 - m}{25}$

$$\frac{dt}{dm} = \frac{25}{750 - m}$$

$$t = \int \frac{25}{750 - m} dm$$

$$= -25 \ln(750 - m) + c$$

When $t = 0, m = 300$:

$$0 = -25 \ln(750 - 300) + c$$

$$c = 25 \ln 450$$

$$t = 25 \ln 450 - 25 \ln(750 - m)$$

$$= 25 \ln \left(\frac{450}{750 - m} \right)$$

$$\frac{t}{25} = \ln \left(\frac{450}{750 - m} \right)$$

$$e^{\frac{t}{25}} = \frac{450}{750 - m}$$

$$750 - m = 450e^{-\frac{t}{25}}$$

$$m = 750 - 450e^{-\frac{t}{25}}$$

iii. Method 1

As $t \rightarrow \infty$:

$$m \rightarrow 750 - 450e^{-\infty} = 750$$

$$\text{Concentration} \left(\frac{m}{V} \right) \rightarrow \frac{750}{50} = 15 \text{ g/L}$$

Method 2

$$\text{Concentration} \rightarrow \frac{dm}{dt}(\text{in}) = 15 \text{ g/L}$$

9. Calculus, EXT1 C3 EQ-Bank 9

$$\frac{dh}{dt} = -Ae^{-0.2t}$$

$$h = \int -Ae^{-0.2t} dt$$

$$= 5Ae^{-0.2t} + C$$

As $t \rightarrow \infty, 5Ae^{-0.2t} \rightarrow 0$

i.e. h eventually stabilises at C .

Find A :

When $t = 0, h = 4$

$$4 = 5Ae^{-0.2t} + C$$

$$4 = 5Ae^0 + C$$

$$A = \frac{4 - C}{5}$$

When $t = 1.5, h = 3$

$$3 = 5 \times \frac{4 - C}{5} e^{-0.2 \times 1.5} + C$$

$$3 = \frac{4}{e^{0.3}} - \frac{C}{e^{0.3}} + C$$

$$\frac{3e^{0.3} - 4}{e^{0.3}} = C \left(1 - \frac{1}{e^{0.3}} \right)$$

$$\frac{3e^{0.3} - 4}{e^{0.3}} = C \left(\frac{e^{0.3} - 1}{e^{0.3}} \right)$$

$$C = \frac{3e^{0.3} - 4}{e^{0.3}} \times \frac{e^{0.3}}{e^{0.3} - 1}$$

$$= 0.1417 \dots \text{ m}$$

$$= 14 \text{ cm (nearest cm)}$$

$\therefore$ Height of fuel in the tank will stabilise at 14 centimetres.

$$\frac{dv}{dx} = c \left(\frac{u^2 - v^2}{v} \right)$$

$$\frac{v}{u^2 - v^2} dv = c dx$$

$$\int \frac{v}{u^2 - v^2} dv = \int c dx$$

$$-\frac{1}{2} \ln(u^2 - v^2) = cx + C$$

When $x = 0, v = 0$:

$$-\frac{1}{2} \ln u^2 = c \times 0 + C$$

$$C = -\ln u$$

$$-\frac{1}{2} \ln(u^2 - v^2) = cx - \ln u$$

$$\ln(u^2 - v^2) = -2cx + 2\ln u$$

$$\ln(u^2 - v^2) = -2cx + \ln u^2$$

$$u^2 - v^2 = e^{-2cx + \ln u^2}$$

$$u^2 - v^2 = e^{-2cx} \cdot e^{\ln u^2}$$

$$u^2 - v^2 = u^2 e^{-2cx}$$

$$v^2 = u^2 - u^2 e^{-2cx}$$

$$= u^2 (1 - e^{-2cx})$$

$$v = \pm u (1 - e^{-2cx})^{\frac{1}{2}}$$

$$= u (1 - e^{-2cx})^{\frac{1}{2}}, \quad (v > 0)$$

$$\frac{dP}{dt} = 0.1P \left(\frac{C-P}{C} \right)$$

$$\frac{dt}{dP} = \frac{10}{P} \left(\frac{C}{C-P} \right)$$

$$= 10 \times \frac{C}{P(C-P)}$$

$$= 10 \left(\frac{1}{P} + \frac{1}{C-P} \right)$$

$$t = 10 \int \frac{1}{P} + \frac{1}{C-P} dP$$

$$= 10 [\ln P - \ln(C-P)] + c$$

$$= 10 \ln \left(\frac{P}{C-P} \right) + c$$

When $t = 0, P = 150\,000$:

$$0 = 10 \ln \left(\frac{150\,000}{C-150\,000} \right) + c$$

$$c = -10 \ln \left(\frac{150\,000}{C-150\,000} \right)$$

$$= 10 \ln \left(\frac{C-150\,000}{150\,000} \right)$$

When $t = 20, P = 600\,000$:

$$20 = 10 \ln \left(\frac{600\,000}{C-600\,000} \right) + 10 \ln \left(\frac{C-150\,000}{150\,000} \right)$$

$$2 = \ln \left(\frac{600\,000}{C-600\,000} \times \frac{C-150\,000}{150\,000} \right)$$

$$2 = \ln \left(\frac{4C-600\,000}{C-600\,000} \right)$$

$$e^2 = \frac{4C-600\,000}{C-600\,000}$$

$$e^2(C-600\,000) = 4C-600\,000$$

$$e^2 C - 4C = e^2 \cdot 600\,000 - 600\,000$$

$$C(e^2 - 4) = 600\,000(e^2 - 1)$$

$$C = \frac{600\,000(e^2 - 1)}{e^2 - 4}$$

$$\approx 1\,131\,121$$

$\therefore$ Carrying capacity $\approx 1\,130\,000$

◆ Mean mark 45%.

i. $V = \pi\left(Rh^2 - \frac{h^3}{3}\right)$

$$\frac{dV}{dh} = \pi(2Rh - h^2)$$
$$\frac{dV}{dt} = k(2R - h) \quad (\text{given})$$
$$\frac{dh}{dt} = \frac{dV}{dt} \cdot \frac{dh}{dV}$$
$$= k(2R - h) \cdot \frac{1}{\pi} \cdot \frac{1}{h(2R - h)}$$
$$= \frac{k}{\pi h}$$

ii. $\frac{dt}{dh} = \frac{\pi h}{k}$

$$t = \int \frac{dt}{dh} dh$$
$$= \frac{\pi}{k} \int h dh$$
$$= \frac{\pi}{k} \left[\frac{h^2}{2} \right] + c$$

When $t = 0, h = 0$

$$\Rightarrow c = 0$$
$$t = \frac{\pi h^2}{2k}$$

Tank is full at time T when $h = R$:

$$T = \frac{\pi R^2}{2k} \text{ seconds}$$

iii. Net water flow $= k(2R - h)$ (given)

Flow in $= 2kR$ (given)

Flow out $= k(2R - h) - 2kR = -kh$

$$\frac{dh}{dt} = \frac{-kh}{\pi h(2R - h)} = \frac{-k}{\pi(2R - h)}$$
$$\frac{dt}{dh} = \frac{-\pi(2R - h)}{k}$$
$$\int k dt = -\pi \int (2R - h) dh$$
$$kt$$

$$= -\pi\left(2Rh - \frac{h^2}{2}\right) + c$$

When $t = 0, h = R$:

$$0 = -\pi\left(2R^2 - \frac{R^2}{2}\right) + c$$
$$c = \pi\left(\frac{3R^2}{2}\right)$$

Find t when $h = 0$:

$$kt = -\pi(0) + \pi\frac{3R^2}{2}$$
$$t = \frac{3\pi R^2}{2k}$$
$$= 3 \times \frac{\pi R^2}{2k}$$

$\therefore$ Tank takes 3 times longer to empty than fill.

◆ Mean mark (ii) 41%.

◆◆ Mean mark (iii) 20%.